



Glove for health

A new wearable device christened MusicGlove can help stroke patients regain their motor skills <http://digit.in/MuGlove>



Closer to Big Bang

IBM shows off some unusual computer systems that could help the world's largest radio telescope to peer 13 billion years back in time. <http://digit.in/IBMBigBang>

Space Age

The idea was very exciting of course since it called for a massive and versatile probe that could not only travel very fast but be very maneuverable as well. This was before the International Space Station had even been conceived.

The Daedalus probe would require 50,000 tonnes of fuel and carry 500 tonnes of scientific payload across a distance of 5.9 light years on a 50 year journey to Barnard's Star. The deuterium/helium-3 powered fusion engine designs called for a top velocity of .12c which is 12 percent the speed of light to achieve its flight plan. The problem of obtaining vast quantities of fuel was solved by siphoning it from nearby planets such as Jupiter. This would not only reduce launch mass but also make fuel easier to obtain since helium-3 is so scarce on Earth. However as the plan was developed further it started requiring features well beyond human technology like self-replicating factories and on-board robot maintenance workers. The final cost came to around USD 100 trillion, which today would translate to over USD 350 trillion dollars.

Laser-powered spacecraft

The biggest problem for long distance voyages has always been time and how to reduce it. But reducing time by going faster isn't as easy as it seems since escaping the Earth's gravity is only the start of the battle. Designing engines that can thrust spacecraft in the vacuum of space is a whole other problem. One of the most interesting and weird ideas involves the use of lasers, and that definitely catches our attention.

The idea for using photonic lasers comes from the Y. K. Bae Corporation which is all about green energy. According to their designs a laser mounted launch platform orbits the Earth and shoots off high intensity laser beams at spacecrafts on their voyage. It's just that simple. While the laser platform and the spacecraft themselves need to be placed in orbit using conventional means, once in space they can easily be propelled using laser bursts from a distance. The laser platform is able to position itself with on board thrusters as the spacecraft moves towards its destination. Thrust is

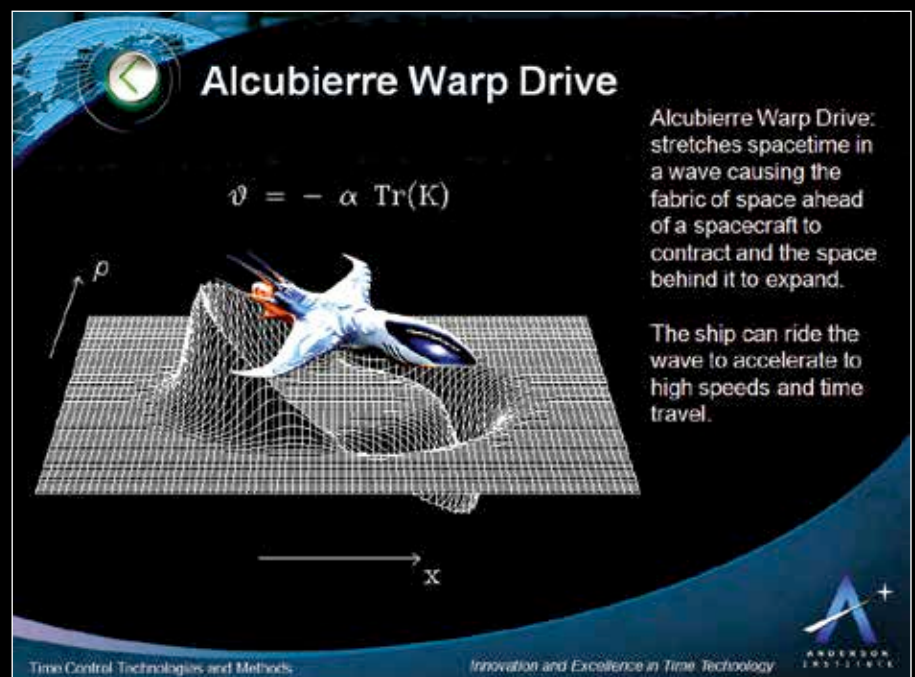
gained by firing streams of photons to increase the craft's velocity. While the plan seems to make sense on paper, the power required to power these orbital lasers is a bit on the higher side at around 1 gigawatt. Not to mention the problems of precision when aiming the laser at increasingly greater distances. And even though both these issues have solutions in the pipeline we just think the idea is just crazy enough to work. Don't look directly into the laser beam through.

Alcubierre drive

This isn't the first time we have looked at the Alcubierre drive. It's a source of con-

anything to move faster than the speed of light. Such a spacecraft would then be static within a "warp bubble" which would be riding along the spacetime curvature as the Alcubierre drive warped the space around it. Far out.

Scientists predict that such a drive could be possible at substantially less power requirements than previously thought and even laid out a basic design. The Alcubierre drive would require the spaceship to be shaped like a tapered oval in the form of a prolate spheroid attached to a large encircling donut like ring. The donut would be composed of exotic matter that would be controlled



This design is literally how the Warp Drive in Star Trek would work. Unbelievable.

stant wonderment and giggles when you realise it's an actual honest to goodness "warp drive". A warp drive that is exactly as Star Trek demonstrated. A curious case of science fiction technobabble that was re-built using cold hard science. Miguel Alcubierre is the scientist who proved this possibility using Einstein's field equations in 1994. He arrived at a solution where instead of a spacecraft moving at light speed, it was space that curved around it. The idea of the Alcubierre drive was then well within the limits of Einstein's theory of general relativity which normally doesn't allow for

to cause space-time to curve around the starship as it was suspended in a warp bubble. This concept if implemented would allow the spacecraft to travel at nearly 10 times the speed of light - making the journey to the Moon in one twelfth of a second, to Pluto in 33 minutes and to Alpha Centauri in 159 days. Of course this technology would require the harnessing of elusive exotic particles and have a host of side-effects, from complete lack of steering control to a sonic boom shockwave that would destroy anything in front of the spacecraft at its destination. **d**